9. Discrete and categorical data analysis: Goodness-of-fit

We have already considered the dependence of discrete measurements (e.g. Poisson, binomial data) to covariates through GLMs.

We will now consider the **goodness-of-fit** of theoretical distributions to observed *discrete* and *categorical* data.

Categorical data arise whenever counts (as opposed to measurements) are made. Subjects (sample items) are classified as belonging to one of a set of categories and the numbers in the categories (the frequencies) are recorded.

Example 9.1 (discrete data)

Accidents of 414 workers over a 3 month period.

Number of accidents	0	1	2	3	4	5	≥ 6	Total
Frequency observed	296	74	26	8	4	6	0	414

Example 9.2 (categorical data)

Eye colours of males visiting an optician, in four categories.

Colour	A	B	C	D
Frequency observed	89	66	60	85

Example 9.3 (categorical data)

In a genetic plant breeding experiment a total of 1,500 plants were categorised into one of four classes, with the following results.

Class	C1	C2	C3	C4
Frequency	1071	62	68	299

Both examples 9.2 and 9.3 are examples of *one-way classifications* of categorical data.

Goodness-of-fit

We usually think of the category as the explanatory variable and the distribution of the items among the categories as the response variable.

Classifications can be regarded as one of:

1. qualitative (no structural relation between categories)
2. ordered
3. quantitative

Classifications with only two categories (male/female; yes/no; on/off) are called dichotomous or binary. In this case, the distinctions (1)–(3) above make no difference to the analysis.

Goodness-of-fit

We can test the goodness of fit of various models (theoretical distributions) to such data using the χ^2 g-o-f test considered previously (for continuous data).

In general we consider the probability p_i of observations to belong to category i, and we have the following hypotheses:

$$\begin{aligned} H_0 &: p_i = \pi_i \text{ for } i = 1, 2, \ldots, r \\ \text{versus } H_1 &: p_i \neq \pi_i \text{ for some } i \end{aligned}$$

for some theoretical probabilities $\pi_1, \ldots, \pi_r$

Goodness-of-fit

We use the usual (Pearson) χ^2 statistic

$$X^2 = \sum_{i=1}^{r} \frac{(o_i - e_i)^2}{e_i}$$

Under H_0 its value should come from a
χ^2 **distribution with** $r - p - 1$ **df**

Here

$$\begin{aligned} o_i &= \text{observed frequencies} \\ e_i &= \text{expected frequencies} \\ r &= \text{number of 'cells' (categories)} \\ p &= \text{number of parameters estimated} \end{aligned}$$

As before, we reject H_0 for large values of the X^2 statistic.

Goodness-of-fit

As with continuous data, caution is required with sparse data leading to small expected frequencies.

Remember that in practice we require $e_i \geq 5$

An alternative rule often used is to ensure that:

$e_i \geq 1$ for all cells, and

$e_i \geq 5$ for at least about 80% of the cells.

If not, **combine adjacent cells** sensibly.

Example 9.1 (continued)

Accidents of 414 workers over a 3 month period.

Number of accidents	0	1	2	3	4	5	≥ 6	Total
Frequency observed	296	74	26	8	4	6	0	414

This is a *quantitative* classification. We want to investigate the fit of a Poisson distribution.

$$\begin{aligned} H_0 &: \text{data come from a Poisson distribution} \\ \text{versus } H_1 &: \text{Poisson not appropriate} \end{aligned}$$

Note that the parameter of the distribution is NOT specified in H_0

Example 9.1

First compute expected frequencies using $\Pr(X = x) = \dfrac{\mathrm{e}^{-\lambda}\lambda^x}{x!}$ where λ is estimated from the data as $\hat{\lambda} = 0.47343$.

Number of accidents, x	0	1	2	3	4	5	≥ 6	Total
Observed frequency, o_x	296	74	26	8	4	6	0	414
Expected frequency, e_x	257.9	122.1	28.9	4.6	0.54	0.05	0.004	414

We need to combine cells to achieve $e_x \geq 5$:

Number of accidents, x	0	1	2	≥ 3	Total
Observed frequency, o_x	296	74	26	18	414
Expected frequency, e_x	257.9	122.1	28.9	5.2	414

Calculate $X^2 = 56.87$, and under H_0 should be from χ^2 distribution with $4 - 1 - 1 = 2$ df. p-value $= 4.5 \times 10^{-13}$

Therefore, we have overwhelming evidence against the null hypothesis.

Example 9.2 (continued)

Eye colours of males visiting an optician, in four categories.

Colour	A	B	C	D
Frequency observed	89	66	60	85

Classification here is *qualitative*.

Does the eye colour of males visiting this optician belong to categories A, B,C or D with equal probabilities?

$$\begin{aligned} H_0 &: p_i = 0.25, \quad i = 1, 2, 3, 4 \\ \text{versus } H_1 &: p_i \neq 0.25 \text{ for some } i \end{aligned}$$

The theoretical model (distribution) is *fully specified* by H_0

Example 9.2

Theoretical frequencies are easy to compute $(300 \times 1/4)$ and we obtain:

$X^2 = 8.03$ which under H_0 should be from χ^2 distribution with $4 - 1 = 3$ df

p-value $= 0.045$

Therefore, we have some evidence against the hypothesis that there is the same chance for each eye colour.

Example 9.3 (continued)

In a genetic plant breeding experiment a total of 1,500 plants were categorised into one of four classes, with the following results.

Class	C1	C2	C3	C4
Frequency	1071	62	68	299

We now wish to test the fit of the following model:

Class	C1	C2	C3	C4
Probability	$(2+\theta)/4$	$(1-\theta)/4$	$(1-\theta)/4$	$\theta/4$

where θ is an unknown parameter such that $0 < \theta < 1$

$$H_0 \quad : \quad \text{above model is appropriate for the observed data}$$

$$\text{versus } H_1 \quad : \quad \text{model not appropriate}$$

Example 9.3

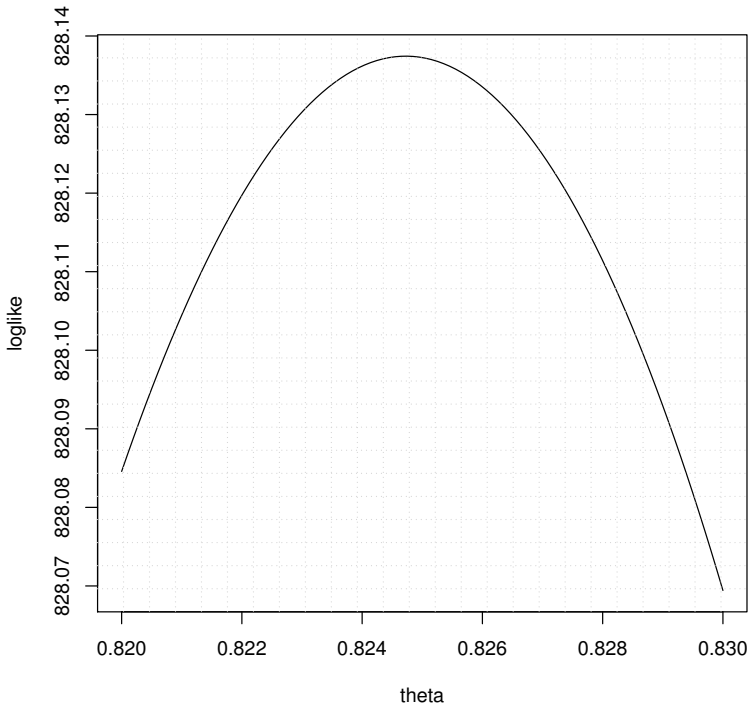
The model is not fully specified by H_0 and we need to estimate the parameter θ

We will use MLE. Log-likelihood is:

$$l(\theta) = 1071 \log(2+\theta) + 130 \log(1-\theta) + 299 \log(\theta) + c$$

In this case an *analytical solution* can be found (differentiate etc – try this as an exercise). However we will demonstrate how the MLE can be found using the plot of $l(\theta)$ in R. First plot $l(\theta)$ on an appropriate range of θ. Then identify in R the θ value that corresponds to the maximum.

Example 9.3



Plot shows that MLE is given by $\hat{\theta} = 0.8247$

Note that other, more formal numerical methods can also be used here – e.g. Newton-Raphson.

Example 9.3

Compute expected frequencies.

Class	C1	C2	C3	C4
Frequency o_x	1071	62	68	299
Expected frequency e_x	1059.3	65.7	65.7	309.3

$X^2 = 0.761$ which under H_0 should be from χ^2 distribution with $4 - 1 - 1 = 2$ df

p-value $= 0.684$

No evidence at all against the hypothesis that plants are classified according to the suggested model.

Goodness-of-fit

Note that in `R`, the command

`chisq.test(x, p = probs)`

where `x` are observed frequencies and `probs` are theoretical probabilities, can **sometimes** be used for a g-o-f test.

However, two important limitations:

- It does not automatically deal with small e_i. This will need to be checked by the user and appropriate action taken.

- It does not allow the number of estimated parameters to be accounted for in d.f.